Chapter 11 CRT

1. Can data in memory be called a file? Explain.
No. Data in memory is stored temporarily in RAM, while a file is a collection of data stored on a storage device, such as a hard drive.

2. Write the import statement required to access the File class in an application.
java
```
import java.io.File;
```

3. Identify the error in the following statement:
java
```
File textFile = new File("c:\inventory.txt");
```
The backslash (\) is an escape character in Java. It must be written as a double backslash (\\).
Correct statement:
java
```
File textFile = new File("c:\\inventory.txt");
```

4. a) Which statement is used to write an exception handler?
The catch statement is used to write an exception handler.
b) Write an exception handler to handle an IOException if a specified file name cannot be used to create a file. The exception handler should display appropriate messages to the user.
java
```
catch (IOException e)
{
   System.out.println("Uh oh spagghetio. There was an error creating the file. ");
   System.out.println("Please check the file name and try again little buddy. ");
}
```

5. a) What is the name of the stream for displaying error messages?
The error stream is called System.err.
b) Where are these messages displayed?
They are displayed on the standard error output, usually the console or terminal window.

6. a) What does the file stream keep track of?
A file stream keeps track of the position in the file where data is being read or written.
b) What characters together make up a line terminator?
A line terminator commonly consists of a carriage return (\r) and a line feed (\n), although the exact terminator depends on the operating system.

7. What two classes are used together to write data to a file?
The FileWriter and PrintWriter classes are used together to write data to a text file.

8. Write a statement to convert account balances that have been read from a text file to a double value and add them to totalBalance.
java
totalBalance += Double.parseDouble(accountBalance);

9. Explain the difference between object serialization and object deserialization.
Serialization: The process of converting an object into a byte stream so it can be saved to a file or transmitted.
Deserialization: The process of reconstructing an object from a byte stream that was previously saved or transmitted.

10. What interface must be implemented if objects of a class are to be written to a file?
The Serializable interface must be implemented.
java
```
import java.io.Serializable;
public class Account implements Serializable
{
   // Class members
}
```